Understanding Diabetes Progression

What patient characteristics go with it, and what can we conclude?

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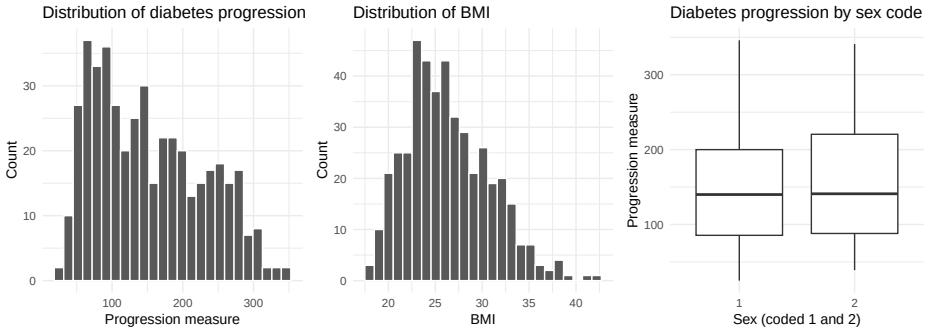
The question and the data

Which patient characteristics go with faster diabetes progression, and what can we safely conclude?

- 442 patients, each measured once at the start
- 10 baseline measures: age, sex, body mass index (BMI), blood pressure and six blood serum measurements
- Outcome: a progression score one year later (higher means more progression)
- No missing values and no duplicate records

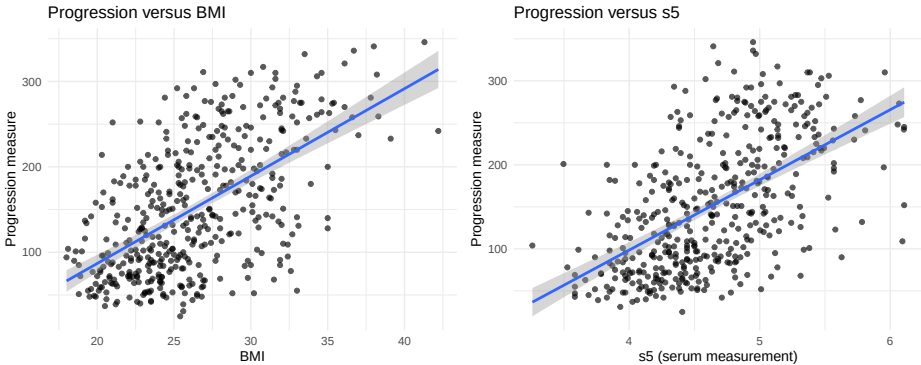
Data: NCSU copy of the diabetes dataset.

What the data look like



- Progression varies a lot: the middle half of patients score between 87 and 212 (full range 25 to 346)
- Most patients have a BMI between 23 and 29; the two sex codes look similar

Which measures go with higher progression?



- Higher BMI and higher s5 (one of the serum measurements) go with higher progression
- Blood pressure also goes with higher progression, but less strongly

What the combined picture shows

Comparing patients who are otherwise similar on the other measures, **on average**:

- Each extra **BMI** point goes with about **6.5** more progression points
- Each extra unit of **s5** goes with about **50** more points (s5 only ranges from 3.3 to 6.1)
- Each extra unit of **blood pressure** goes with about **0.9** more points, a smaller difference
- Together, these measures account for less than half of the differences between patients (about 46% to 48%)

How much can we trust it?

Fairly solid:

- The same pattern appears in both versions of the analysis
- The size of the BMI difference is estimated fairly precisely: between about 5 and 8 points per BMI point

Important cautions:

- Individual patients vary a lot: a typical miss is about 56 points on a scale from 25 to 346
- The fit is weaker for patients with the lowest predicted progression
- The same patients were used to build and check the analysis, with no outside test

What we can and cannot conclude

We can say

- Higher BMI and higher s5 go with higher progression, then blood pressure
- These differences are clear in this group of patients

We should not say

- That lowering BMI or s5 would slow progression
- That we can predict progression for an individual patient
- That the results apply to other groups of patients

These are observational data: other factors we could not see (diet, treatment, how long someone has had diabetes) may explain part of what we found. **We make no causal claims.**

Questions and discussion

Thank you